

Supplementary Figure S1



Figure S1. Visualization of the dataset

Supplementary Figure S2

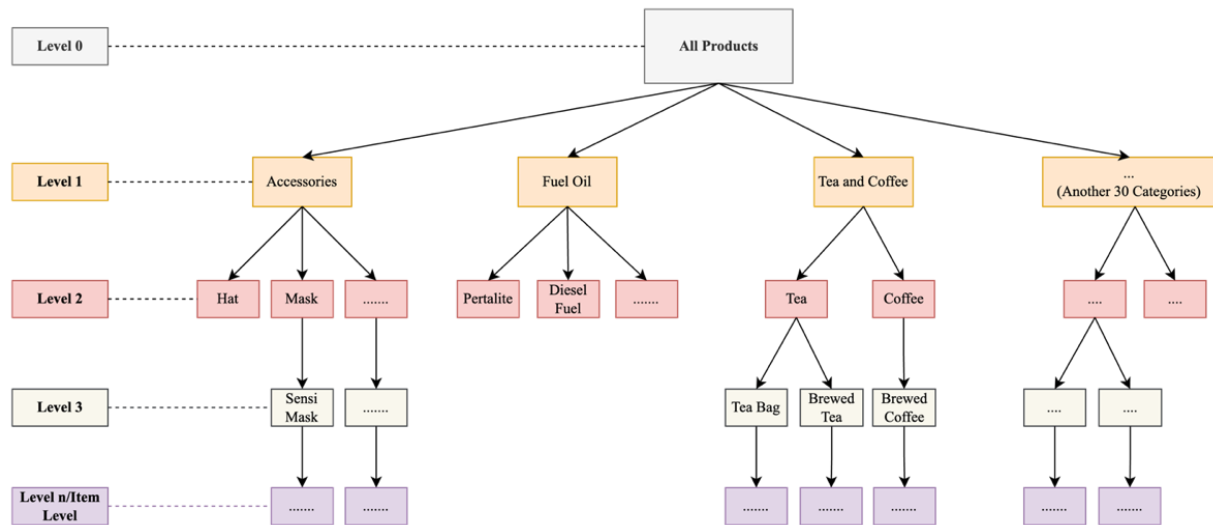


Figure S2. Figure 1 Product Hierarchy
(Source: Griva et al (2018))

Supplementary Figure S3

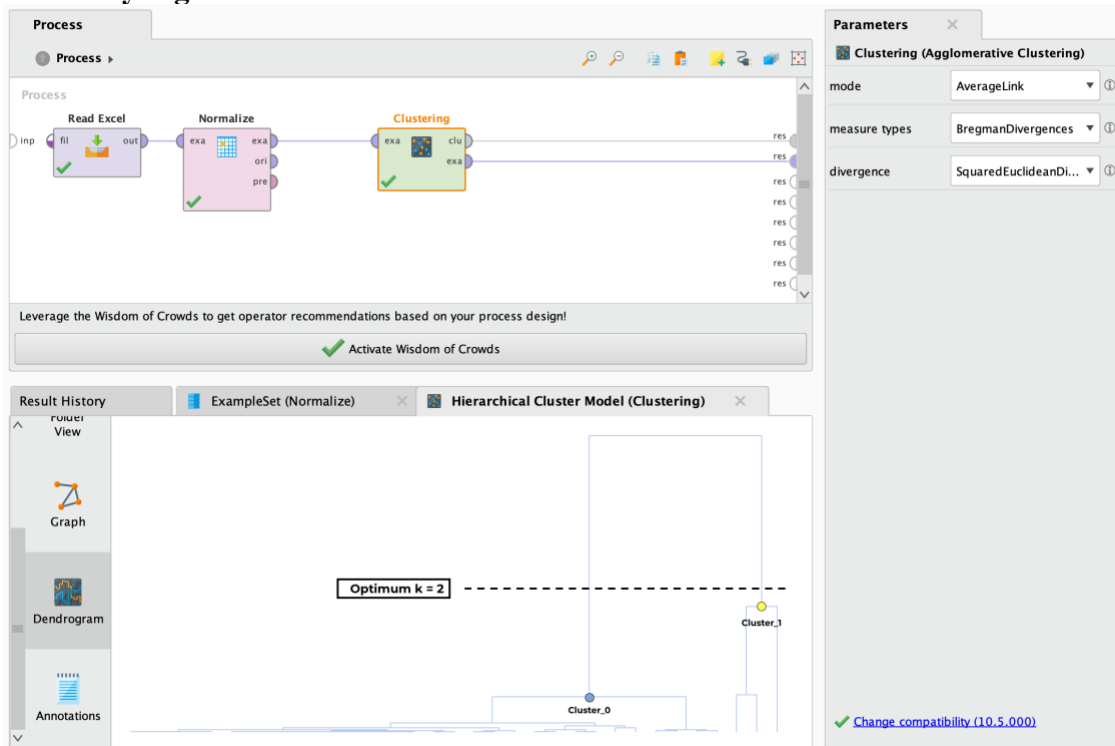


Figure S3. Agglomerative Clustering Model

Supplementary Figure S4

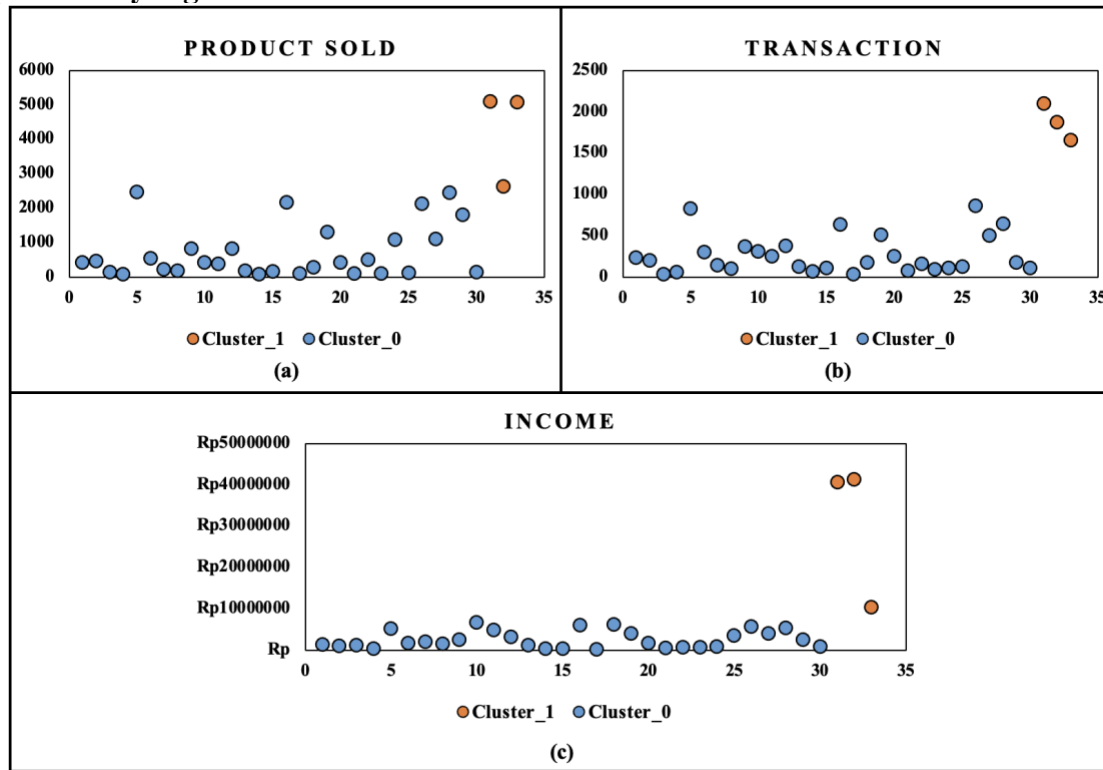


Figure S4. Clustering Output

Axis X represents the products in the following order:

No	Category	No	Category
1	Accessories	18	Oil, Butter & Margarine
2	Stationery	19	Medicine
3	Rice	20	Body Care
4	Bread Ingredients	21	Dental Care
5	Beverage	22	Hair Care
6	Spices	23	Facial Care
7	Diapers & Bandage	24	Plastics
8	Electronic	25	Credit & E-wallet
9	Frozen	26	Breads & Biscuits
10	Water Gallons & LPG	27	Milk
11	Sugar	28	Tea & Coffee
12	Hygiene	29	Eggs
13	Baby Stuff	30	Flour & Pasta
14	Other Necessities	31	Fuel Oil
15	Toys	32	Cigarettes
16	Instant Food	33	Snack & Candy
17	Powder Drink and Syrup		

Supplementary Figure S5

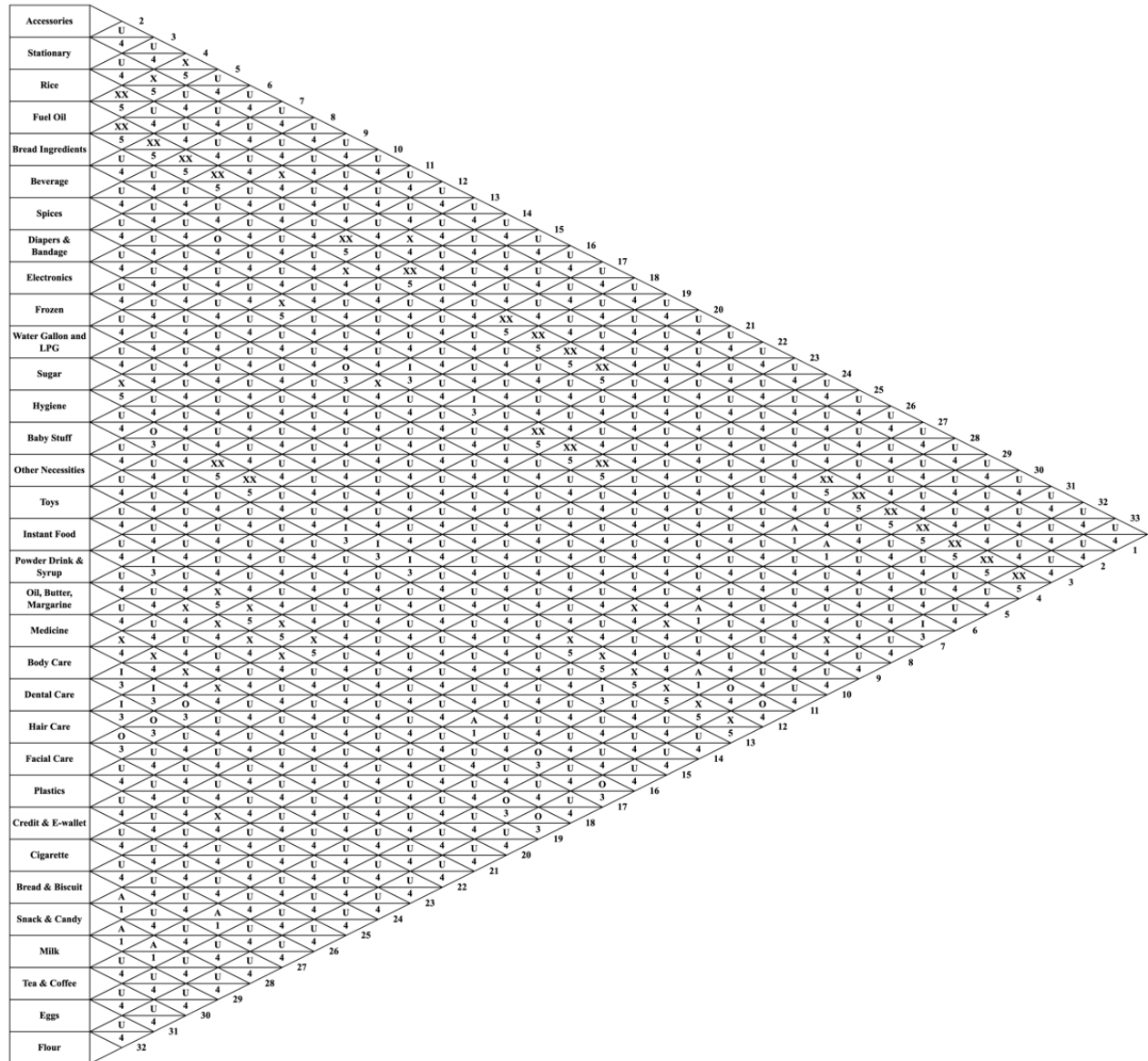


Figure S5. Activity Relationship Chart (ARC) Second Scenario

The diagram illustrates 18 different shelving unit configurations, each labeled from Shelf A to Shelf W. Each shelf has a specific layout of items, with some shelves having multiple rows or columns of items. The items are categorized into various groups like Diapers & Bandage, Instant Food, Spices, Snack & Candy, Bread & Biscuit, etc.

Shelf A

Shelf A			
Diapers & Bandage			
Diapers & Bandage	Other Necessities	Other Necessities	
Hair Care	Dental Care	Body Care	
Instant Food	Hair Care	Dental Care	Body Care
Spices	Hygiene		
Hygiene			
Stock			Hygiene

Shelf B

Shelf B			
Instant Food		Diapers & Bandage	
Instant Food			
Instant Food			
Spices	Powder Drink & Syrup	Oil, Butter & Margarine	Instant Food
Spices			
Flour & Pasta		Instant Food	
Stock			

Shelf C

Shelf C	
Milk	
Tea & Coffee	
Tea & Coffee	
Sugar	

Shelf D

Shelf D	
Baby Stuff	Beverage
Beverage	
Beverage	
Beverage	

Shelf E

Shelf E	
Snack & Candy	
Snack & Candy	
Snack & Candy	
Snack & Candy	
Snack & Candy	

Shelf F

Shelf F	
Snack & Candy	
Snack & Candy	
Bread & Biscuit	
Bread & Biscuit	
Bread & Biscuit	

Shelf G

Shelf G	
Bread & Biscuit	
Bread & Biscuit	
Bread & Biscuit	
Bread & Biscuit	

Shelf H

Shelf H	
Instant Food	
Instant Food	
Instant Food	
Instant Food	

Shelf I

Shelf I	
Bread Ingredients	
Bread Ingredients	
Plastics	
Plastics	

Shelf J

Shelf J	
Snack & Candy	
Snack & Candy	
Snack & Candy	
Snack & Candy	
Snack & Candy	

Shelf K

Shelf K	
Medicine	
Facial Care	
Facial Care	
Baby Stuff	Plastics

Shelf L

Shelf L	
Snack & Candy	
Snack & Candy	
Snack & Candy	
Snack & Candy	

Shelf M

Shelf M	
Milk	
Stationery	
Stationery	
Stationery	

Shelf N

Shelf N	
Cigarettes	

Shelf O

Shelf O	
Electronics	
Electronics	
Electronics	
Electronics	

Shelf P

Shelf P	
Rice	

Shelf Q

Shelf Q	
Frozen	

Shelf R

Shelf R	
Water Gallons	
LPG	

Shelf S

Shelf S	
Accessories	
Accessories	
Accessories	
Accessories	

Shelf T

Shelf T	
Snack & Candy	
Snack & Candy	
Snack & Candy	
Snack & Candy	

Shelf U

Shelf U	
Toys	

Shelf V

Shelf V	
Fuel Oil	

Shelf W

Shelf W	
Tea & Coffee	

Figure S6. Shelf Product Allocation Second Scenario

Supplementary Figure S7

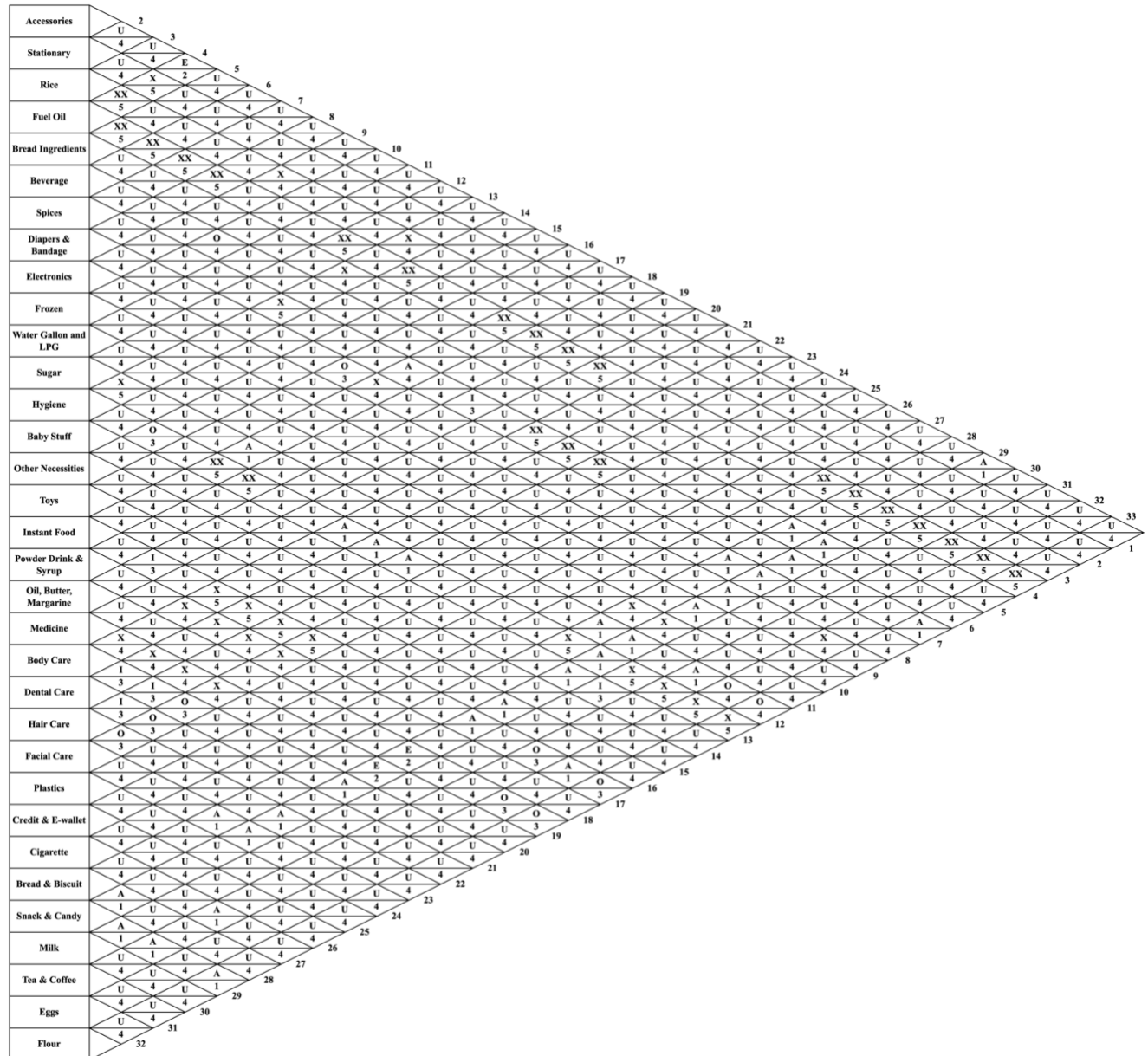


Figure S7. Activity Relationship Chart (ARC) Third Scenario

Supplementary Figure S8

The diagram illustrates 23 shelves (A-W) with the following items:

- Shelf A:** Diapers & Bandage, Diapers & Bandage, Other Necessities, Other Necessities, Body Care, Instant Food, Hair Care, Dental Care, Spices, Hygiene, Instant Food, Hygiene, Stock.
- Shelf B:** Instant Food, Diapers & Bandage, Instant Food, Instant Food, Spices, Powder Drink & Syrup, Oil, Butter & Margarine, Instant Food, Spices, Flour & Pasta, Instant Food, Rice, Stock, Eggs.
- Shelf C:** Milk, Tea & Coffee, Tea & Coffee, Sugar.
- Shelf D:** Baby Stuff, Beverage, Beverage, Beverage, Beverage.
- Shelf E:** Snack & Candy, Snack & Candy, Snack & Candy, Snack & Candy, Snack & Candy.
- Shelf F:** Snack & Candy, Snack & Candy, Bread & Biscuit, Bread & Biscuit, Bread & Biscuit.
- Shelf G:** Milk, Milk, Bread Ingredients, Bread Ingredients.
- Shelf H:** Instant Food, Instant Food, Instant Food, Instant Food.
- Shelf I:** Bread & Biscuit, Bread & Biscuit, Bread & Biscuit, Bread & Biscuit.
- Shelf J:** Snack & Candy, Snack & Candy, Snack & Candy, Snack & Candy, Snack & Candy.
- Shelf K:** Medicine, Facial Care, Facial Care, Baby Stuff, Plastics.
- Shelf L:** Bread & Biscuit, Bread & Biscuit, Bread & Biscuit, Bread & Biscuit.
- Shelf M:** Stationery, Stationery, Stationery, Plastics.
- Shelf N:** Cigarettes.
- Shelf O:** Electronics, Electronics, Electronics, Electronics.
- Shelf P:** (Empty)
- Shelf Q:** Frozen.
- Shelf R:** Water Gallons, LPG.
- Shelf S:** Accessories, Accessories, Accessories, Accessories.
- Shelf T:** Snack & Candy, Snack & Candy, Snack & Candy, Snack & Candy.
- Shelf U:** Toys.
- Shelf V:** Fuel Oil.
- Shelf W:** Tea & Coffee.

Figure S8. Shelf Product Allocation Second Scenario

Supplementary Table S1**Table S1. Attribute Description**

Attribute	Data type	Description
Transaction id	Numerical	Unique code for each transaction
Product	Integer	Types of products purchased
Quantity	Numerical	Quantity of products purchased
Price	Numerical	Product unit price
Total Price	Numerical	Total spending in a transaction

Supplementary Table S2**Table S2. Clustering Dataset Description**

Attribute	Data type	Description
Category	Integer	Product Categories (Level 1)
Income	Numerical	Revenue from each product category
Transaction Frequency	Numerical	Frequency of product category occurrence in a single transaction
Product Sales	Numerical	Number of products from each category sold

Supplementary Table S3**Table S3 Biner Representation Dataset Example**

No	Accessories	Stationery	Rice	Fuel Oil	...
1	0	0	0	1	...
2	0	0	0	1	...
...	...	...	...	...	...
6964	0	0	0	0	...

Supplementary Table S4**Table S4 Degree of Proximity ARC**

Code	Description
A	Absolute necessity to be adjacent
E	Very important to be adjacent
I	Important to be adjacent
O	Adequate or Usual
U	Not important to be adjacent
X	Forbidden to be adjacent

Supplementary Table S5**Table S5 Reasons for Proximity of ARC**

Reason Code	Proximity Reason Description
1	Lift value >1
2	Confidence value >20%
3	Interconnected categories
4	No relationship between categories
5	Categories causing contamination

Supplementary Table S6**Table S6. First Scenario Output Association Rules Model**

No	Antecedent (X)	Consequent (Y)	α	β	Lift Ratio
1	Tea & Coffee	Snack & Candy	3.0%	32.9%	1.386
2	Beverages	Snack & Candy	4.4%	36.8%	1.548
3	Instant Food	Snack & Candy	3.5%	38.0%	1.601
4	Bread & Biscuit	Snack & Candy	5.8%	46.5%	1.956

Supplementary Table S7**Table S7. Second Scenario Output Association Rules Model**

No	Antecedent (X)	Consequent (Y)	α	β	Lift Ratio
1	Bread & Biscuit	Beverages	2.6%	21.0%	1.773
2	Tea & Coffee	Sugar	2.0%	21.8%	6.108
3	Beverages	Bread & Biscuit	2.6%	22.0%	1.773
4	Snack & Candy	Bread & Biscuit	5.8%	24.2%	1.956
5	Tea & Coffee	Cigarrete	2.8%	30.7%	1.141
6	Tea & Coffee	Snack & Candy	3.0%	32.9%	1.386
7	Beverages	Snack & Candy	4.4%	36.8%	1.548
8	Instant Food	Snack & Candy	3.5%	38.0%	1.601
9	Frozen	Snack & Candy	2.1%	39.0%	1.641
10	Milk	Snack & Candy	2.9%	40.5%	1.704
11	Bread & Biscuit	Snack & Candy	5.8%	46.5%	1.956
12	Sugar	Tea & Coffee	2.0%	56.2%	6.108

Supplementary Table S8

Table S8. Third Scenario Output Association Rules Model

No	Antecedent (X)	Consequent (Y)	α	β	Lift Ratio
1	Bread & Biscuit	Beverage	2.6%	21.0%	1.773
2	Sugar	Bread & Biscuit	0.8%	21.3%	1.718
3	Milk	Bread & Biscuit	1.6%	21.4%	1.729
4	Tea & Coffee	Sugar	2.0%	21.8%	6.108
5	Beverage	Bread & Biscuit	2.6%	22.0%	1.773
6	Oil, Butter, Margarine	Snack & Candy	0.6%	22.6%	0.952
7	Diapers & Bandage	Bread & Biscuit	0.5%	23.1%	1.862
8	Flour & Pasta	Spices	0.4%	23.4%	5.419
9	Medicine	Snack & Candy	1.7%	23.6%	0.993
10	Accessories	Snack & Candy	0.8%	23.8%	1.003
11	Spices	Instant Food	1.0%	23.9%	2.607
12	Hygiene	Snack & Candy	1.3%	24.2%	1.019
13	Snack & Candy	Bread & Biscuit	5.8%	24.2%	1.956
14	Electronic	Snack & Candy	0.4%	24.3%	1.022
15	Flour & Pasta	Snack & Candy	0.4%	24.3%	1.024
16	Sugar	Cigarettes	0.9%	24.5%	0.910
17	Hair Care	Snack & Candy	0.6%	24.8%	1.046
18	Spices	Snack & Candy	1.1%	25.9%	1.091
19	Sugar	Instant Food	0.9%	26.1%	2.845
20	Hair Care	Hygiene	0.6%	26.1%	4.837
21	Body Care	Hygiene	1.0%	26.3%	4.866
22	Face Treatment	Bread & Biscuit	0.4%	26.3%	2.124
23	Body Care	Snack & Candy	1.0%	27.8%	1.172
24	Accessories	Fuel Oil	1.0%	28.9%	0.961
25	Face Treatment	Snack & Candy	0.4%	29.5%	1.241
26	Tea & Coffee	Cigarette	2.8%	30.7%	1.141
27	Sugar	Snack & Candy	1.1%	31.3%	1.319
28	Tea & Coffee	Snack & Candy	3.0%	32.9%	1.386
29	Baby Supplies	Snack & Candy	0.6%	34.9%	1.469
30	Beverage	Snack & Candy	4.4%	36.8%	1.548
31	Eggs	Instant Food	0.9%	36.9%	4.018
32	Instant Food	Snack & Candy	3.5%	38.0%	1.601
33	Frozen	Snack & Candy	2.1%	39.0%	1.641
34	Milk	Snack & Candy	2.9%	40.5%	1.704
35	Eggs	Snack & Candy	1.1%	41.9%	1.764
36	Dental Care	Hygiene	0.4%	43.1%	7.974
37	Diapers & Bandage	Snack & Candy	0.9%	43.4%	1.825
38	Bread & Biscuit	Snack & Candy	5.8%	46.5%	1.956
39	Toys	Snack & Candy	0.8%	52.3%	2.204
40	Sugar	Tea & Coffee	2.0%	56.2%	6.108